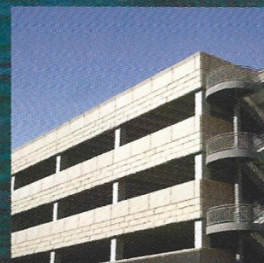




The Chemical Company

Durability Solutions



Master
Builders

Durability Issues and Concrete Solutions



Sulfate Attack

Soil, groundwater, or water bodies containing sulfates can be detrimental to concrete. The chemical reactions cause deleterious expansive forces within the concrete matrix, resulting in cracking and deterioration. **Rheomac SF** silica fume increases the resistance of concrete to sulfate attack by reducing permeability. **Rheocrete 222+** admixture also works to increase sulfate resistance due to its permeability-reducing mechanism.

Alkali-Silica Reactivity (ASR)

Concrete containing certain siliceous reactive aggregates may experience abnormal internal expansion and cracking due to alkali-silica reactivity (ASR).

BASF Admixtures **ASRx™ 30LN** lithium-based durability admixture is specifically formulated to inhibit deleterious expansion in concrete susceptible to ASR because of the unavoidable use of reactive aggregates. ASRx 30LN admixture converts reactive silica in the aggregate to a nonexpansive, nondestructive material.

Rheomac SF silica fume, may also be used in concrete to combat ASR. The use of this pozzolan reduces both concrete permeability and the alkalis that react with the silica in the reactive aggregates.

Cracking

Cracking of concrete is a major concern. Cracking may be due to plastic shrinkage and plastic settlement of concrete in the unhardened state or to drying shrinkage or other factors in the hardened concrete.

Cracking due to plastic shrinkage and plastic settlement can be minimized by adding synthetic fibers, such as **Fibermesh*** from Propex Concrete Systems, to the concrete. Because the fibers are uniformly distributed throughout the concrete mixture, they control bleeding, keep the heavier constituents of the mixture in place, and inhibit cracks. Depending on the application rate, fibers can inhibit up to 80-100% of plastic shrinkage cracking.

BASF Admixtures **Tetraguard® AS20** shrinkage-reducing admixture can be used to reduce drying shrinkage and the potential for subsequent cracking in concrete. Tetraguard AS20 admixture functions by reducing capillary tension of pore water, thereby reducing the internal stresses that cause concrete to shrink. Reducing drying shrinkage lowers the potential for cracking thus improving aesthetics, watertightness and durability. Drying shrinkage may be reduced by as much as 80% at 28 days and up to 50% at one year, depending on the concrete mixture and the dosage of Tetraguard AS20 admixture. Secondary benefits of Tetraguard AS20 admixture include a reduction in compressive creep, reduced curling and carbonation.

Strength

For many concrete structures, strength is an important property that affects production, quality, and durability. Supplementary cementitious materials are often essential to the production of high-strength concrete. **Rheomac SF** silica fume is a micro-filling material that physically and chemically fills the voids between cement particles thereby increasing concrete strength at all ages. The use of BASF Admixtures **Glenium®** and **Rheobuild®** families of high-range water-reducing admixtures in combination with Rheomac SF silica fume will further increase strength. In specialized applications, strengths can be increased to 20,000 psi (138 MPa) and higher.

Durability Issues and Concrete Solutions



Durability and service life are major considerations in the design of concrete structures subject to aggressive environments. Typical projects requiring long-term durability include dams, bridges, parking garages, water and wastewater treatment facilities, marine and various other structures.

Designing more durable, longer-lasting concrete structures is a constant goal of engineers, architects, owners, government agencies, contractors and concrete producers worldwide.

For nearly a century, BASF Admixtures, Inc. has developed the most technologically advanced concrete admixtures to make these goals a reality. Today, it is common for concrete structures to be designed for service lives in excess of 75 years.

The durability of concrete is affected by many issues including:

- Corrosion
- Sulfate Attack
- Alkali-Silica Reactivity
- Cracking
- Strength
- Economic Benefit

Corrosion

Chloride ions enter concrete from deicing salts on roads and bridges or from seawater in marine environments. Other sources of chloride ions include admixtures made with intentionally-added calcium chloride, chloride-contaminated aggregates and/or mixing water and salts in ground water. Chloride ions promote corrosion of reinforcing steel and other metals in concrete which then expand and cause the surrounding concrete to crack and deteriorate.

The **Rheocrete®** family of corrosion inhibiting admixtures offers options to delay the onset and reduce the rate of corrosion of steel reinforced concrete structures.

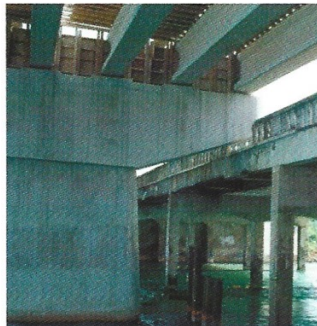
Rheocrete 222+ admixture

- Organic-based amine-ester corrosion inhibitor
- Reduces concrete permeability
- Forms a protective film at the steel surface

Rheocrete CNI admixture

- 30% calcium nitrite
- Inorganic anodic-type corrosion inhibitor
- Reacts with ferrous ions to form an oxide film at the steel surface

Further durability enhancement against corrosion is attainable by combining Rheocrete corrosion inhibitors with the use of **Rheomac® SF** silica fume. Rheomac SF silica fume reduces concrete permeability by providing additional hydration products that reduce the number and size of capillary pores. This makes it even more difficult for chloride ions to penetrate the concrete's surface.

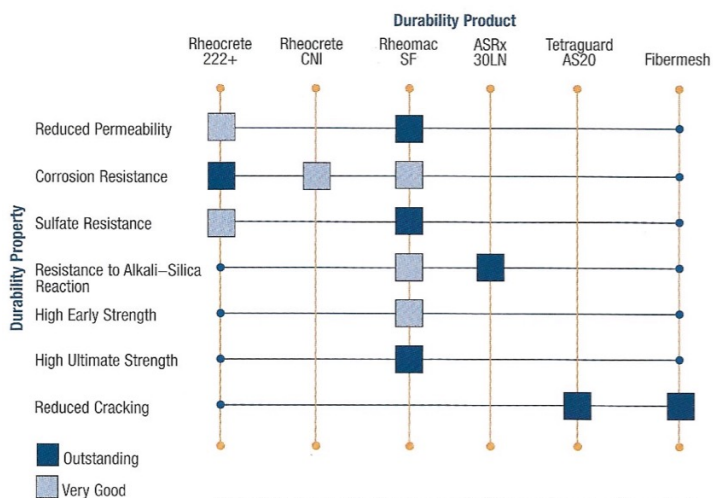
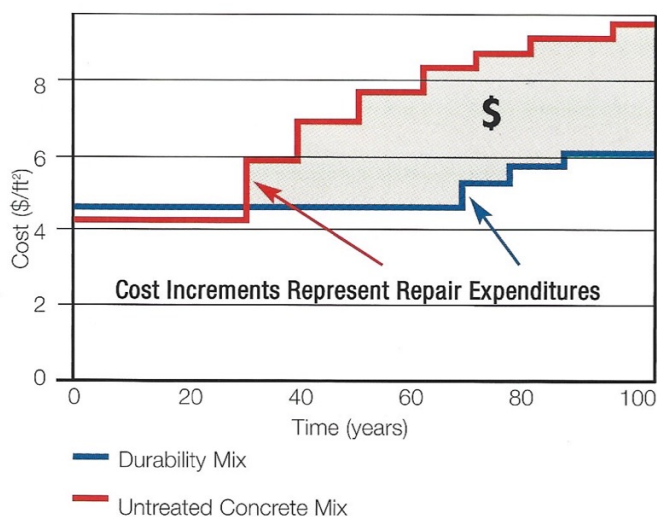


The bridge on the right is experiencing heavy corrosion. The bridge on the left was built to prevent corrosion by using high performance concrete with durability admixtures. Durability issues, such as corrosion found on roads, bridges, and parking structures, can be greatly improved with BASF Admixtures durability admixtures.



Economic Benefit

Concrete designed for durability will typically increase the initial cost of the mixture. However, the use of BASF Admixtures durability admixtures can provide significant economic benefits over the life of a structure as illustrated in the following chart.



This Admixture Performance Guide can be used to select the optimum BASF product(s) to improve specific durability properties.

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BASF Admixtures, Inc. is a leading provider of innovative chemical admixtures and silica fume used in the ready mix, precast, manufactured concrete products, underground construction and paving markets in the United States and Canada. The company's respected Master Builders brand products are used to improve the placing, pumping, finishing, appearance and performance characteristics of concrete.